Water Pump Final Report

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MAE 2250: Mechanical Synthesis

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TABLE OF CONTENTS

Section I: Water Pump Design Overview and Process …….. 2

Section II: Images and Renders ……. 8

Section III: Fabrication Plan …….. 10

Section IV: Graphs, Charts, Tables …… 14

Appendix …… 19

A: Part Drawings …… 19

B: FEA Analysis …… 30

C: Team Charter …… 34

D: Sketches, Notebook Drawings …… 37

Section I: Water Pump Design Overview and Process

After choosing a piston pump, because of its simplicity and reliability, as our ideal pump, we decided on two modifications to the simple pump to improve performance. First, we chose to design a double acting piston pump since it pumps water on both strokes instead of just one, thus improving the efficiency of the pump. We knew this would be more challenging to manufacture and prevent leaking, but we wanted to try it regardless. Second, we chose to make two pumps instead of one. Due to the small size of the given cylinders, we were unsure that just one double acting pump would suffice to pump one liter in a minute and we thus decided to utilize our budget to the best of our ability and create a double-acting, double piston pump.

Our pump functions by means of a Scotch-yoke mechanism, intaking water through the inlet valves and then pushing the water out on the opposite strokes. Using one motor shaft, this mechanism translates the rotational movement of the motor to linear motion that pushes the pump heads in the pumps. The Scotch-yoke mechanism utilizes a slot to engage the pin rotating with the motor switch between forms of motion.



Figure 1.1 Scotch-yoke mechanism

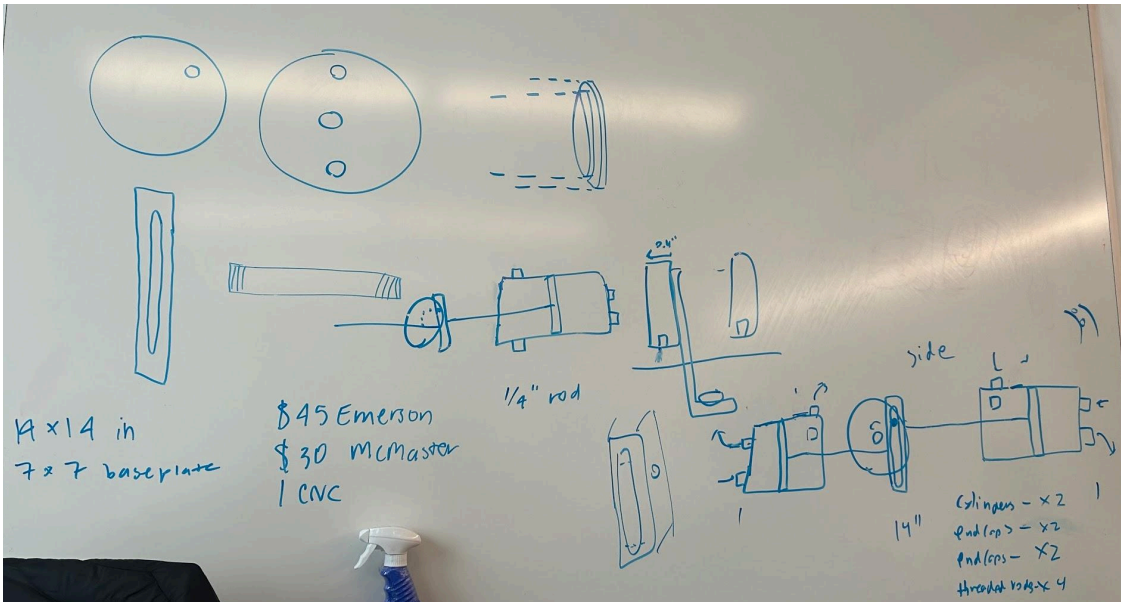


Figure 1.2 Initial white-board sketches during our first design meeting

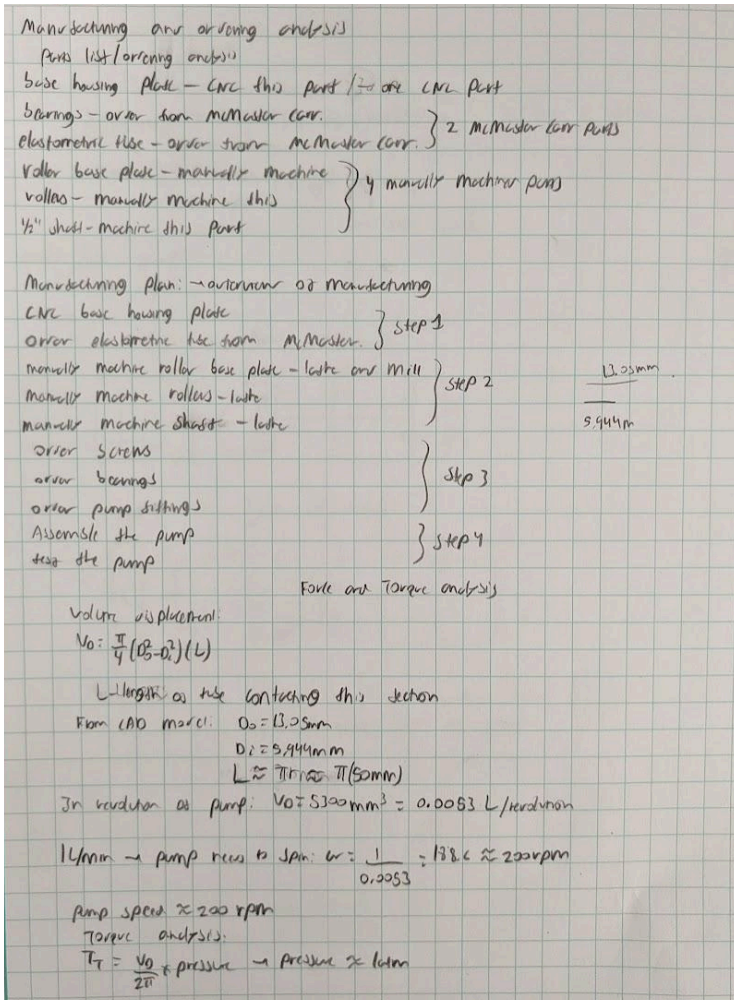


Figure 1.3 Notebook sketches

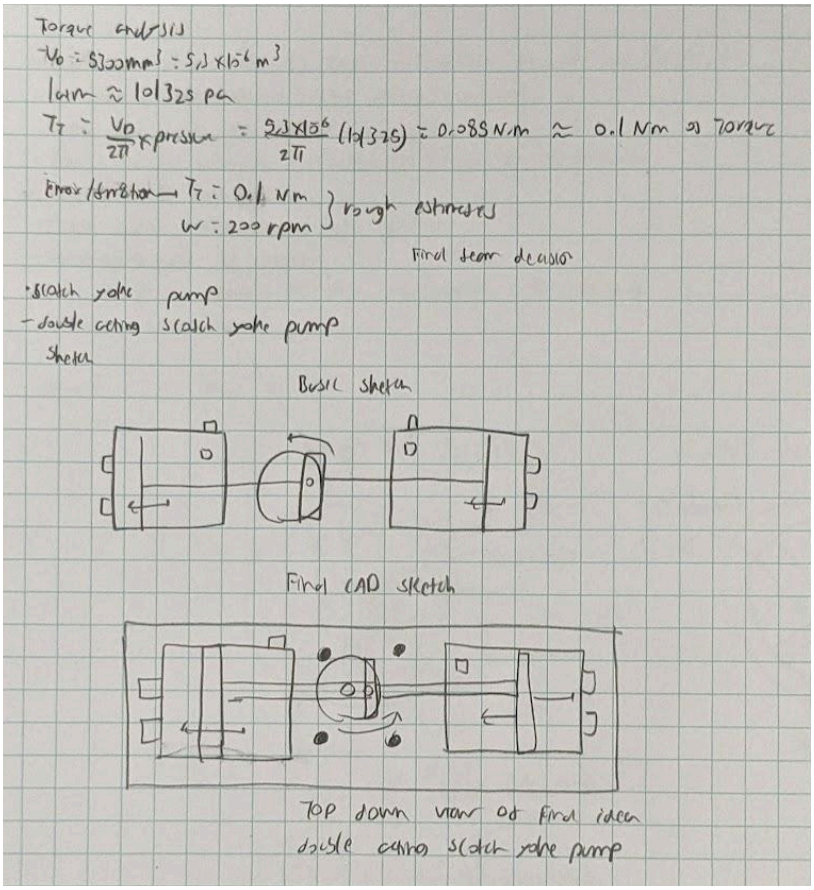


Figure 1.4 More notebook sketches

After completing the design process and creating a functioning CAD model in Fusion 360, we then began the machining process. Since the majority of our parts were from the premade Emerson shop parts, we didn't have much difficulty with our budget due to the affordability of these, and were able to start machining relatively quickly. With a total of 6 different parts to machine, we divided them amongst the group and had pretty much no issues with making them. They ranged from being as simple as cutting a rod to size to adding a slot to a piece of stock.

One issue we ran into was the question of how to manufacture angle brackets to attach our pumps to the baseplate. In the end, we opted to 3D print them, since that would be strong enough and machining those would be difficult. When we ran our pump on the final test day at the higher speed, these pieces ended up snapping when things got misaligned, but for slower speeds, the 3D printed pieces worked very well. Given more time, we would have come up with a stronger, easier to assemble design for this.

Once all of the parts were machined, we assembled and tested our pump. Initially, we had some tolerance issues which was causing our pump to get stuck at a very specific spot in its rotation. To resolve this issue, we sanded some parts down, applied grease to our pump to reduce friction, and ran it with a drill to smooth down and stuck points. Another small issue we had was that our pump wasn't designed for assembly as well as it could have been, which resulted in there being only one order to assemble the pump in. During our initial assembly and troubleshooting session, this was an inconvenience because we were taking it apart and putting it together a lot to solve our tolerance issues. This wasn't a large issue and didn't affect us much overall. We would also definitely redesign certain portions of our pump to make it easier to assemble if given more time.

During the day of water pump testing, we ended up pumping 10.5 liters of water in a minute, which was close to, but slightly under our calculated 12 L/min. This was with our pump being run at the slower of the two speed options. When we increased the speed to the higher option, our pump got slightly misaligned, snapping the aluminum rods and mounting brackets. This could have been prevented by stronger materials or better alignment of our pieces from better tolerances.

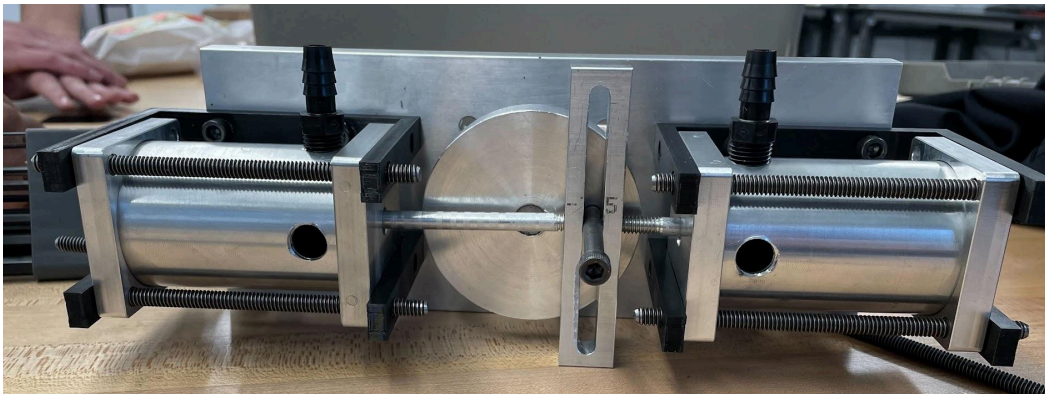


Figure 1.5 Closeup of pump during initial assembly and testing

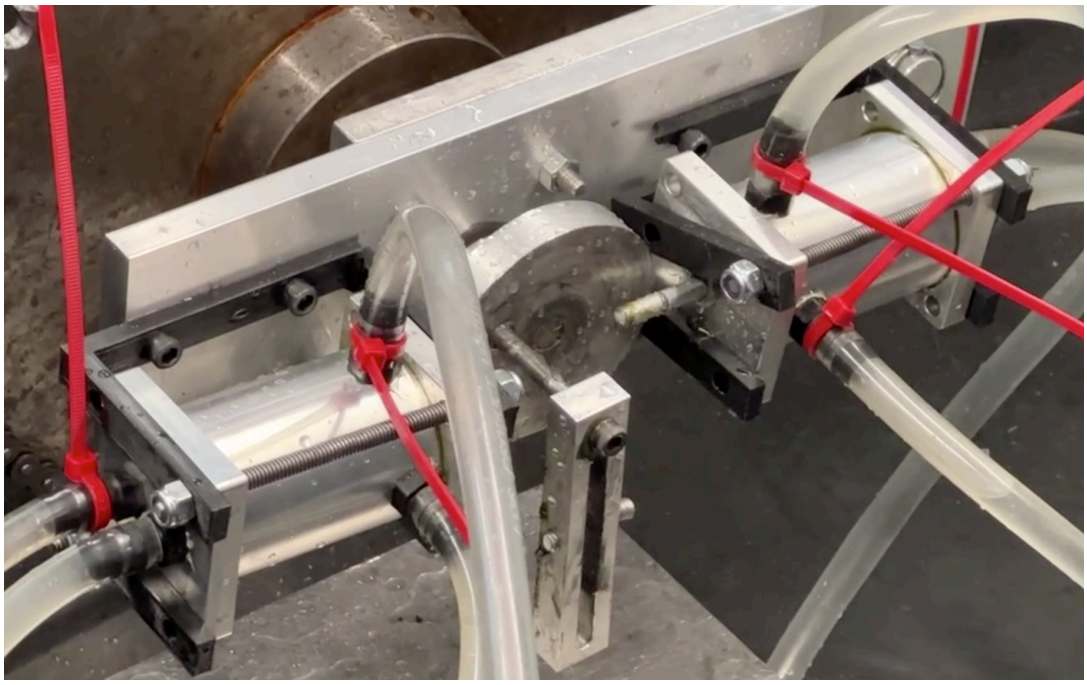


Figure 1.6 After running pump at a higher RPM, the aluminum rod and angle brackets snapped

Section II: Images and Renders

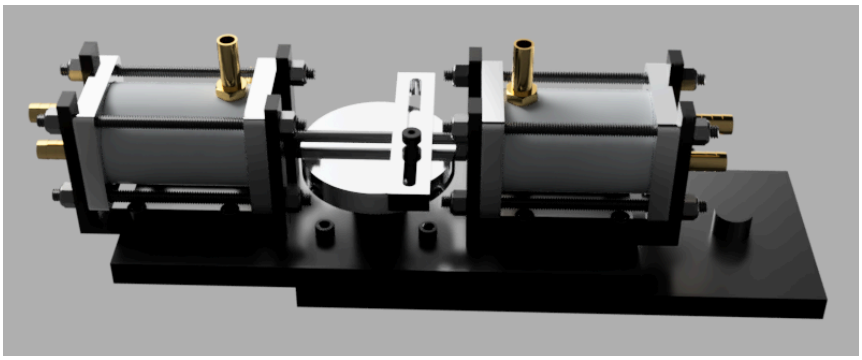


Figure 2.1 Rendered image of our final CAD model

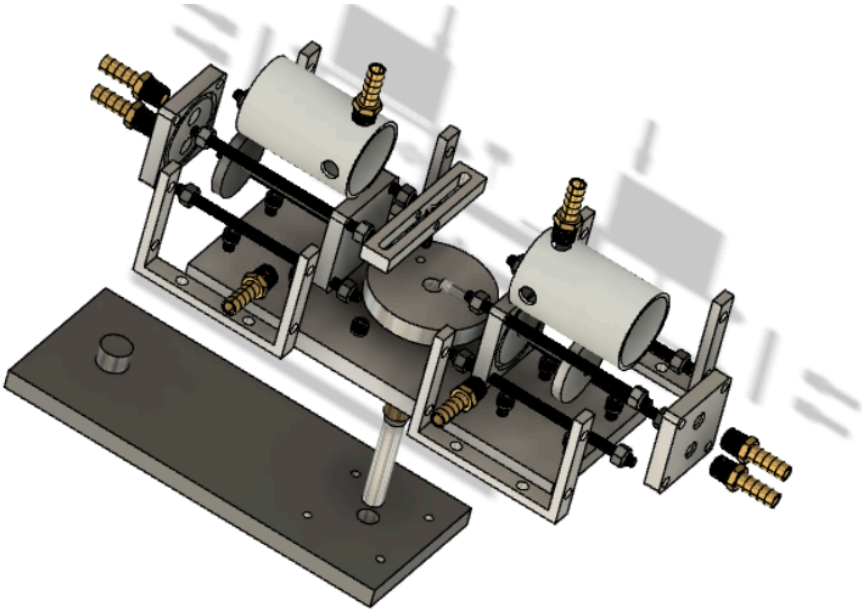


Figure 2.2 Exploded view of our CAD model

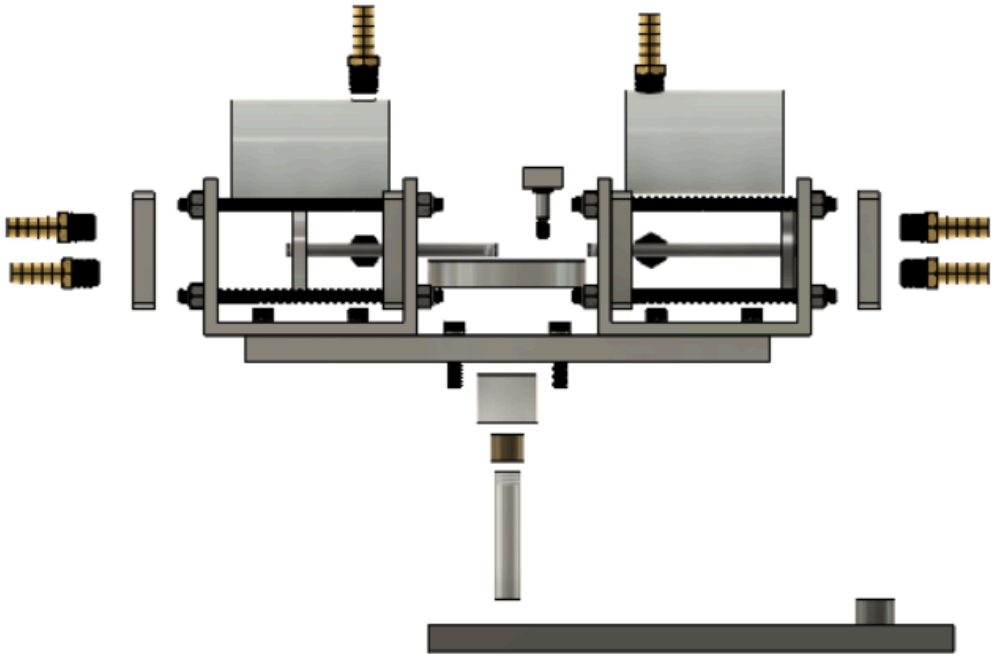


Figure 2.3 Exploded view of the CAD model from a different angle

Section III: Fabrication

Our fabrication process began by listing all of the necessary components for our pump design. Throughout the ideation and CAD process, we kept a record of the parts within our folder in Fusion 360. Based on our list of parts, we decided which should be purchased or fabricated in the machine shop, as detailed in Table 3.1. The table also lists who machined what parts

Manufacturing	**Ordering**	**Machining Responsibility**
Base Plate - Mill	12 x ¼”-20 Screws	Albert
Link - Mill	8 x ¼” threaded rod	Leilani and Nyra
2 x Piston Heads - Lathe	8 x ¼”-20 Locknuts	Aravind and Rafael
2 x Rods - Lathe	2 x 2 hole End Caps	Abdul and Rafael
Turning Circle - Mill/Lathe	2 x 1 hole End Caps	Aravind
Drive Shaft - Lathe	2 x Cylinder	Aravind and Albert
Knob - Lathe	8 x Hose fitting	Nyra and Leilani
4 x Angle Brackets	N/A	3D Printed
Spacer	Machine Shop Stock	Aravind

Table 3.1 Manufacturing/Ordering parts list

Since using standardized fasteners was both cost-effective and time efficient, all of our screws and nuts were to be purchased from McMaster-Carr. The piston housings would have been very difficult to manufacture ourselves and would have added unnecessary variability to our designs. Because of this, we agreed to take advantage of the prefabricated cylinders and end caps from the Emerson Machine Shop as well as the hose fittings available on McMaster-Carr. The remaining parts were manufactured in the machine shop from stock also purchased through this vendor.

Our base plate, link, and turning circle were made using the milling machine since they included holes that seemed easiest to make using vertical drilling methods, though the diameter of the turning circle also needed to be shaved down to size on the lathe. The pistonheads, rods, driveshaft, and knob were all then fabricated on the lathe since we could find stock in our desired shapes and then shave them down to the sizes that our design required. Our last manufactured components were the angle brackets, which we were originally unsure of how to manufacture but decided to have them 3D printed since that was the easiest and cheapest solution.

After our manufacturing and ordering analysis, we created a timeline to plan out the fabrication of our product. Our timeline included which slots (mill vs. lathe) in the machine shop we could register for:

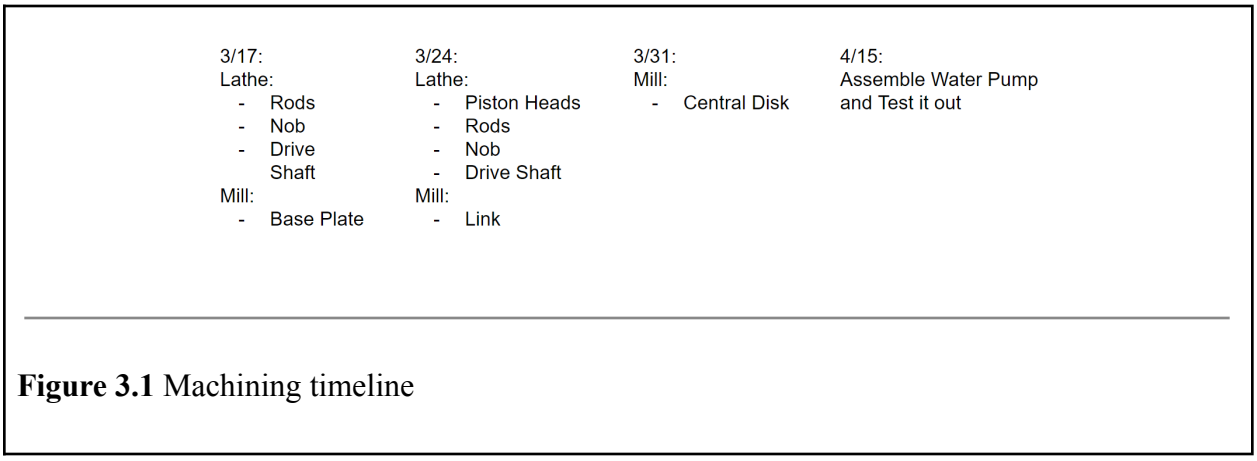


Figure 3.1 Machining timeline

To actually fabricate the parts, we made part drawings for each machined part. All of the part drawings that we made can be found in part A of the appendix. The tolerances for our parts were 0.005” for hole placements, 0.003” for edge dimensions and lengths, and 0.001” for holes.

In our design, the lengths of parts and the placement of holes were very crucial to its success. If the rods were too long or the piston heads were too thick, the pump could get jammed and possibly break. The sizes of the holes were also crucial to ensure that leaking was minimized. The holes in the end caps, in particular, had to be very tight to ensure the best possible seal. As a result of this, they had the tightest tolerances. Finally, the relative placement of the holes was also important to ensure that our pump would fit together. However, since the hole placement was not crucial to the success of the pump, we had a looser tolerance than for other dimensions.

Thus, we had to have very tight, but slightly different, tolerances on our parts. Furthermore, we also ran FEA analysis on our design to ensure that the cylinders and rods would not break under expected loading conditions. We ran simulations on these parts because they were the most likely to break under the operation of the pump. These simulations are also attached in part B of the appendix.

Concluding our fabrication process, the final cost of our project is summarized in the bill of materials included below:

Bill Of Materials - Emerson

Item Name	Associated CAD Part	Quantity	Length (in)	Cost for 1 (or 1 in)	Total Cost	Total	Budget
End Cap with 2 Holes	end cap with 2 holes	2		$1.00	$2.00	$32	$45
End Cap with 1 Hole	end cap with 1 hole	2		$1.00	$2.00		
1/4" Threaded Rod	threaded rod	1	24	$0.09	$2.04		
1/4" Rod	rod	2	3.75	$0.10	$0.75		
1/4" Rod	turning circle part	1	0.75	$0.10	$0.08		
Cylinder	cylinder	2		=$1.00	$2.00		
Base (4"x0.5"x10")	base	1	10	$1.18	$11.80		
1/4 - 20 Screws	Mounting	4		$0.16	$0.64		
1/4 - 20 Screws	Mounting	8		$0.17	$1.36		
1/2" Rod	drive shaft	1	2.5	$0.23	$0.58		
2" Diameter x 3" Aluminum Rod	pump head	2	0.25	$1.00	$0.50		
1/4 - 20 Lock Nuts	Mounting	8		$0.06	$0.48		
Nylon pipe fittings (3/8" barbed x 1/4"NPT)	Pipe fittings	8		$0.55	$4.40		
1/2" ID x 5/8" OD x 1/2" depth lubricated bronze bushings	Central Bushing	1		$0.60	$0.60		
1/4" x 1" (max length 10") (base)	Angle Brackets	2	10	$0.14	$2.80		

Figure 3.2 Bill of Materials for Emerson parts

Bill of Materials/Cost Analysis

McMaster	Emerson	Misc.	Total
$3.22	$32	$0	$35.24

Item Name	Part Number	URL	Cost	Quantity	Total Cost	Total Cost	Budget
Multipurpose 6061 Aluminum, 1.5" Diameter	1610T13	https://www.mcmaster.com/1610T13-1610T107/?s=aluminum+disk	$3.22	1	$3.22	$3.22	30

Figure 3.3 Overall bill of materials

Section IV: Water Pump Ideation and Efficacy Analysis

Although there are many pump configurations capable of satisfying the 1 L/min throughput outlined in the design objectives of the pump, careful consideration was taken to analyze the unavoidable truths of reality that could sabotage the completion of this project.

First and foremost, our team analyzed the machinability and manufacturability of the various proposed pumps. Gear pumps were the first to be considered as impractical as they were impractical to machines (metal gears were desirable due to the high RPM required by gear pumps to be effective) and ineffective at pumping low-viscosity fluids such as water.

A familiar system was also valued over exotic pump methods, and due to the limitations of pump knowledge (from not having taken Fluid Mechanics), the centrifugal pump was also deemed unsuited towards our team's success. Impeller vanes and the volute casing design would have to rely on intuition rather than calculated design due to time and knowledge constraints.

Similarly, the peristaltic pump was also considered to be unsuitable for us. Factors such as tube compression and bearing spacing were all very foreign and it would be unlikely that we would have created a well-working pump by the testing deadline.

Ultimately, we settled on the piston pump design. A functional breakdown, as shown in Figure 4.2, helped by laying out the required features for the part design; the parts required seemed reasonable to meet. In addition, the variety of configurations and methods relating to machining and operation made room for a lot of possibilities and the basic physical principle of water displacement tailored this design to be more digestible for us. A thorough morphological design analysis, as shown below in Figure 4.2, and following team design meetings culminated

in the design of a Scotch-yoke piston pump design—optimal for its simplicity, cost, and compactness (from the implementation of double-acting chambers).

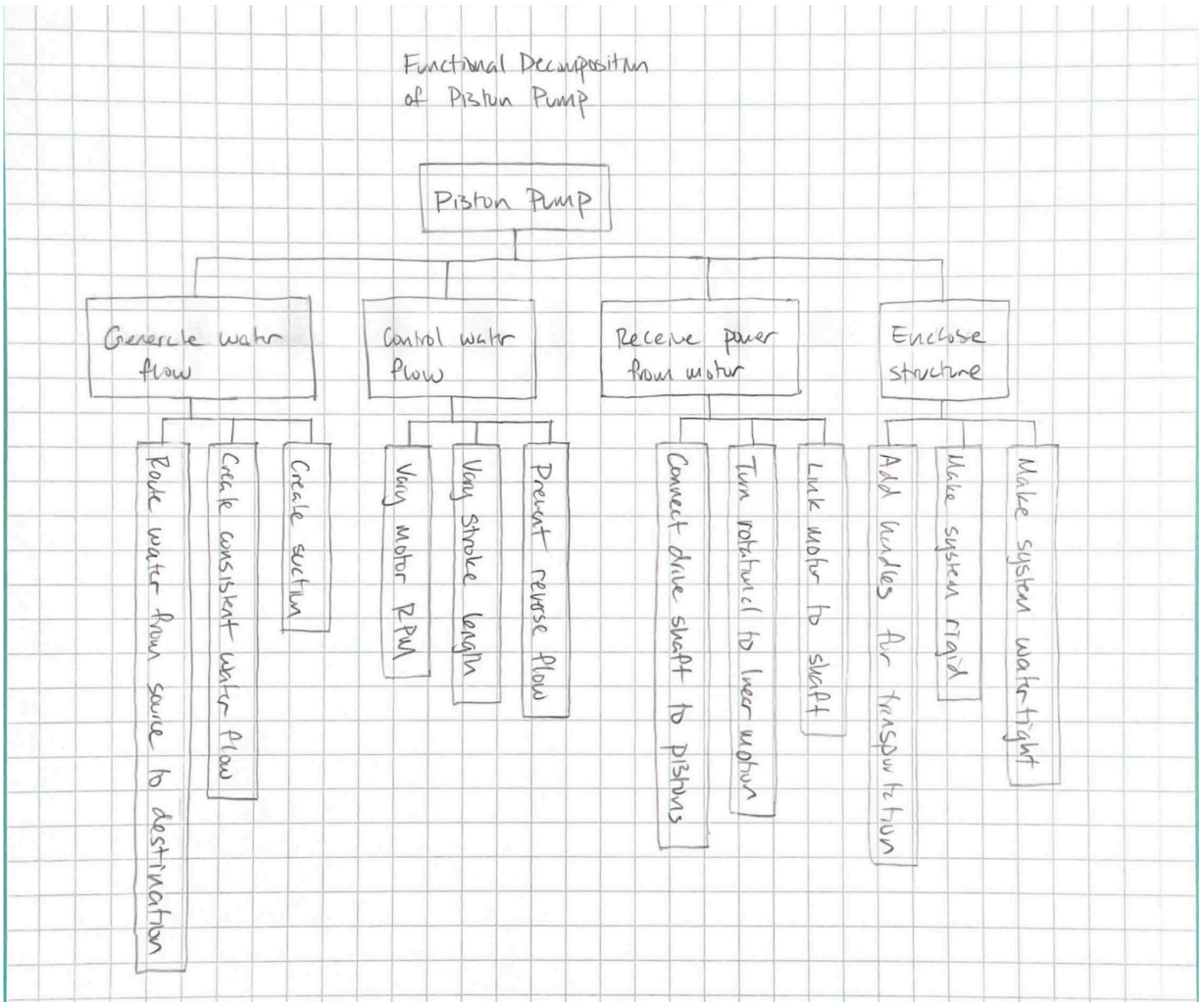


Figure 4.1 Functional decomposition diagram for a piston pump.

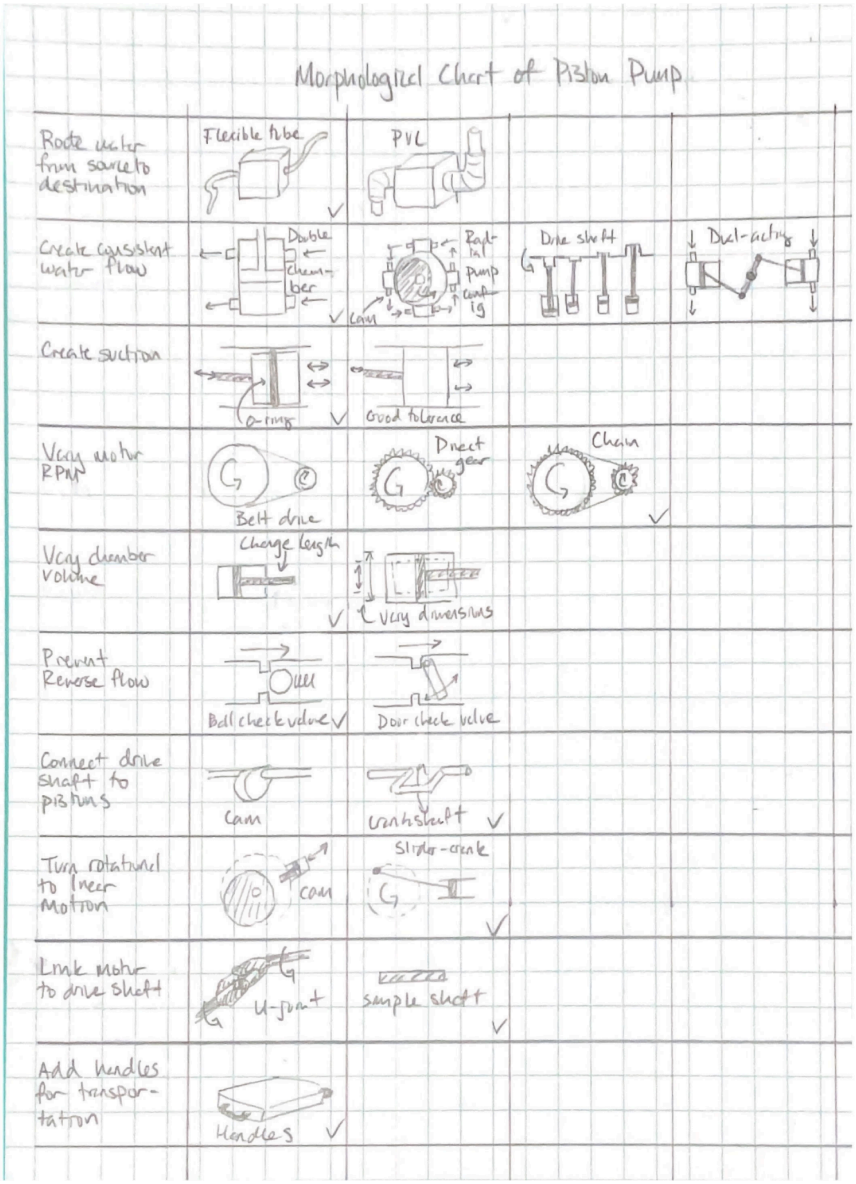


Figure 4.2 Morphological diagram for a piston pump.

To verify the pump's efficacy, pump calculations were made resulting in the formula (a derivation is given below in Figure 4.3):

$$P(x) = \frac{2A_1Rx}{\pi}\left(dgp - \frac{\left(A_2^2 - A_1^2\right)}{3A_2^2}R^2px^2\right)$$

Where P(x) represents the average power draw with respect to angular velocity (x). A_1 and A_2 are the head area and tube cross-sectional area respectively. R is the radius of the central wheel (where the yoke is pinned to the drive links) and d, g, and p represent the height, gravitational constant, and density of the fluid respectively. By settling on a combined mechanical and electrical efficiency of around 55% (which should be low enough to account for a reasonable amount of leakage within the system), each acting chamber was predicted to pump 50 mL/s or 3 L/min. As such, for all 4 chambers connected, we expected 12 L/min.

From the actual test, our pump output 10.5 L/min, which is close to our 12 L/min theoretical prediction (13.3% difference), suggesting that the pump calculations were reasonable and within the error you would expect from hand calculations that did not take into account machining defects and tolerancing issues in our parts.

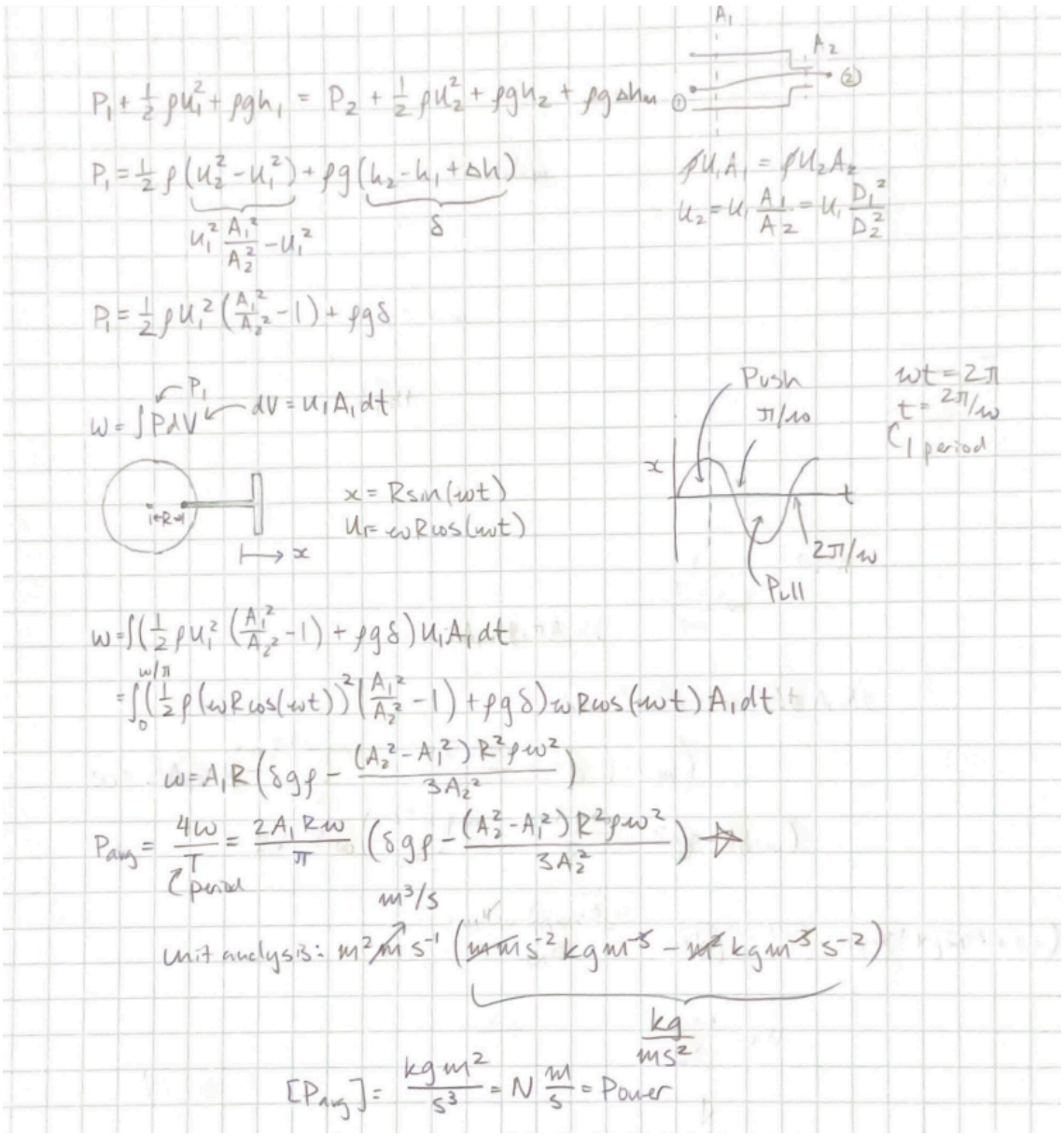


Figure 4.3 Power and output calculations for pump output

Appendix

A. Part Drawings

Note: Many of these parts were Emerson stock that we modified. The part drawings for these parts only show the modifications that we had to make on the parts. The existing features were not dimensioned or toleranced since we didn't need to make them.

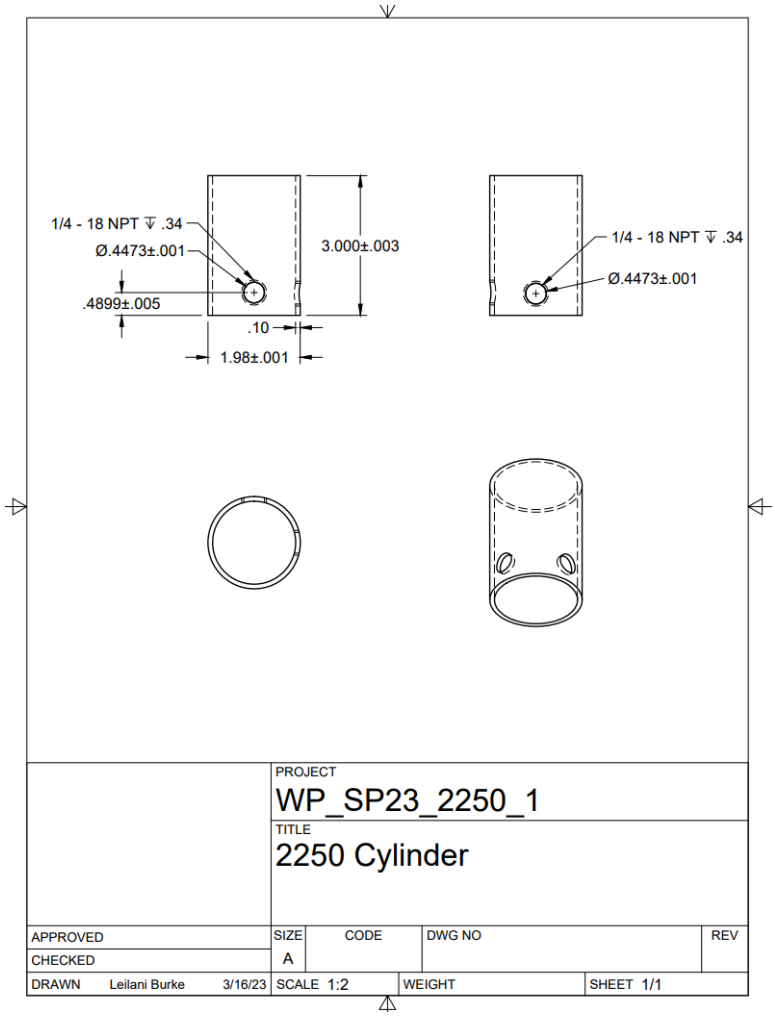


Figure A.1 Part drawing for the cylinders (modified from Emerson stock)

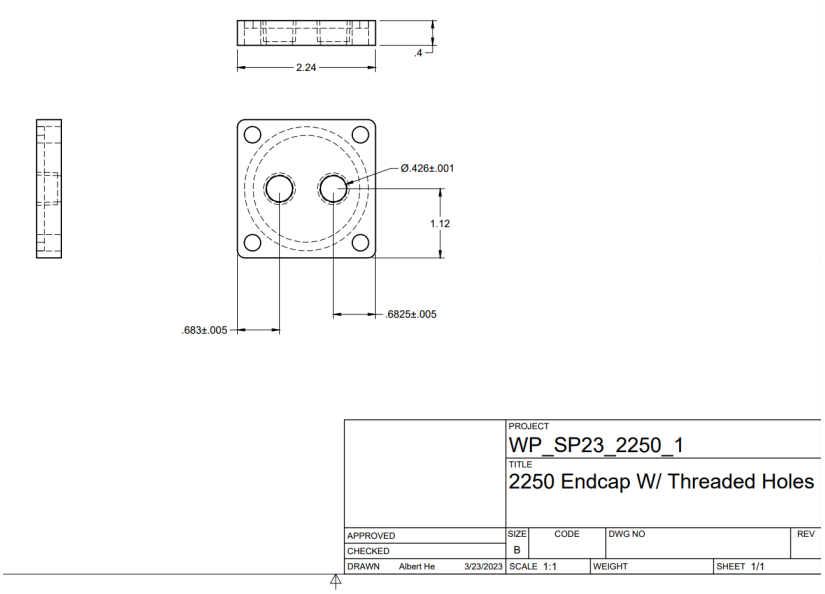


Figure A.2 Part drawing for the end caps (modified from Emerson stock)

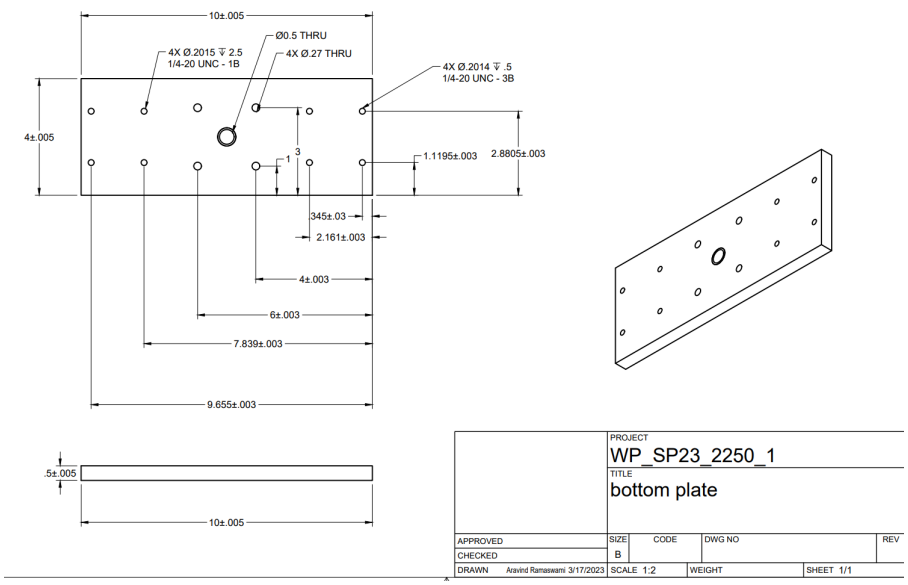


Figure A.3 Part drawing for the machined baseplate

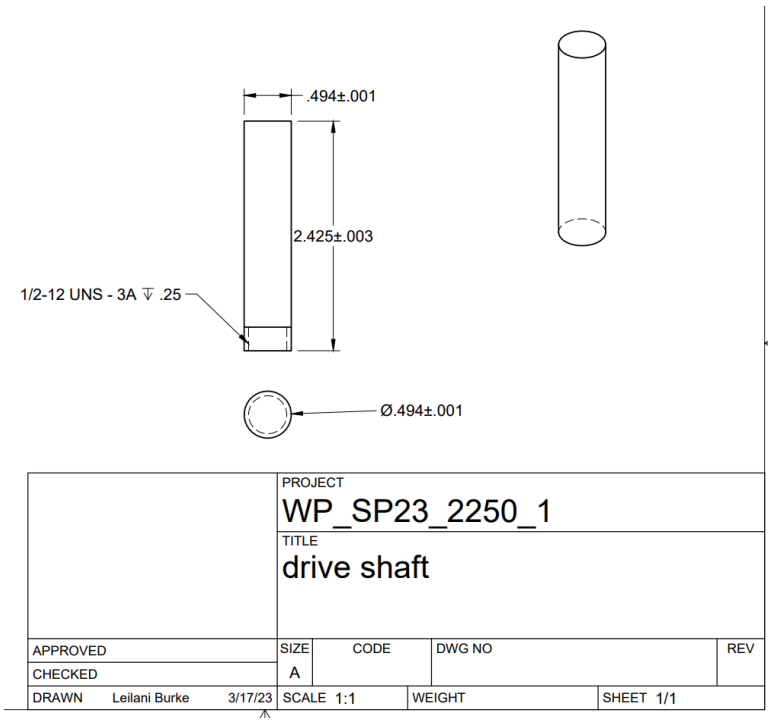


Figure A.4 Part drawing for the machined drive shaft

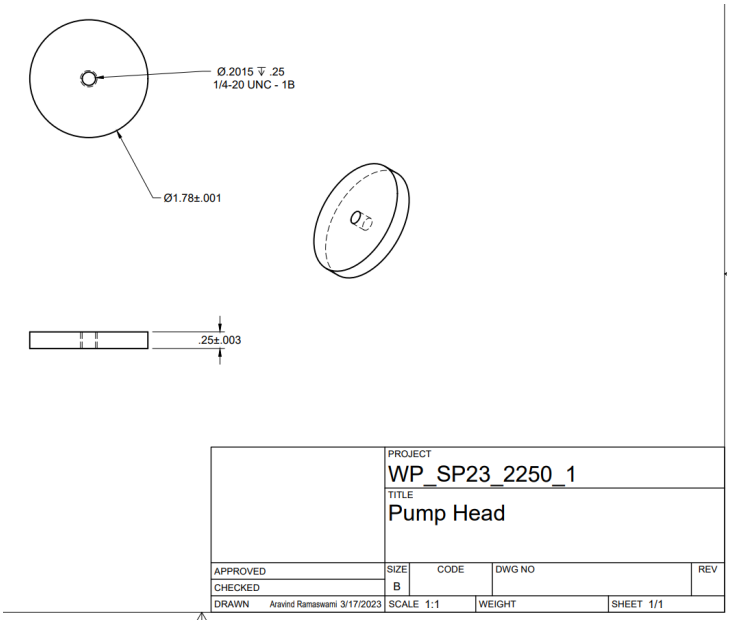


Figure A.5 Part drawing for the machined piston head made from PVC

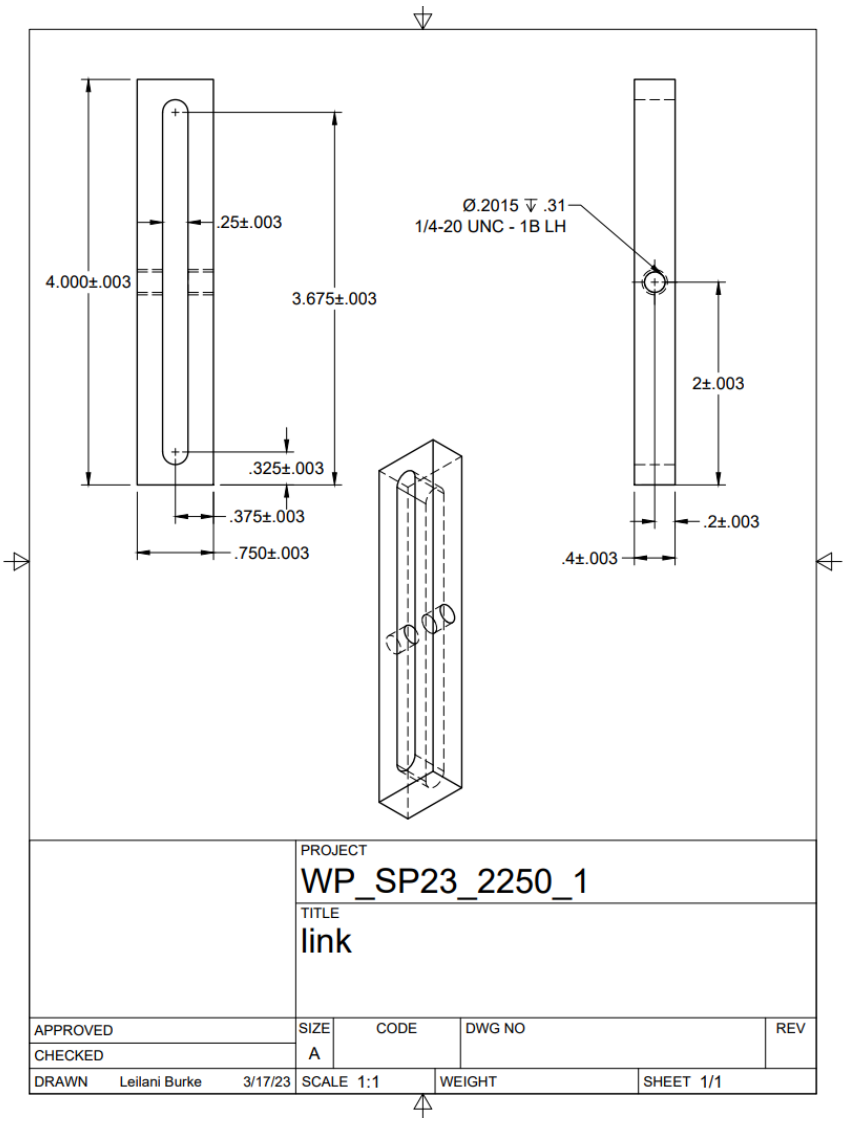


Figure A.6 Part drawing for the machined link with a slot

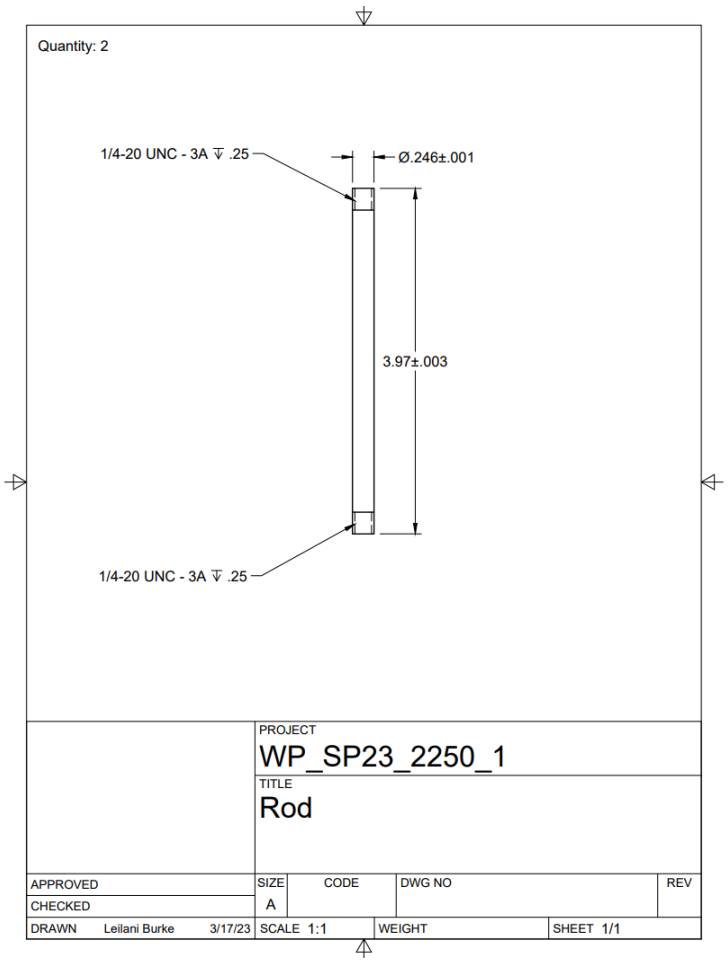


Figure A.7 Part drawing for the machined rods

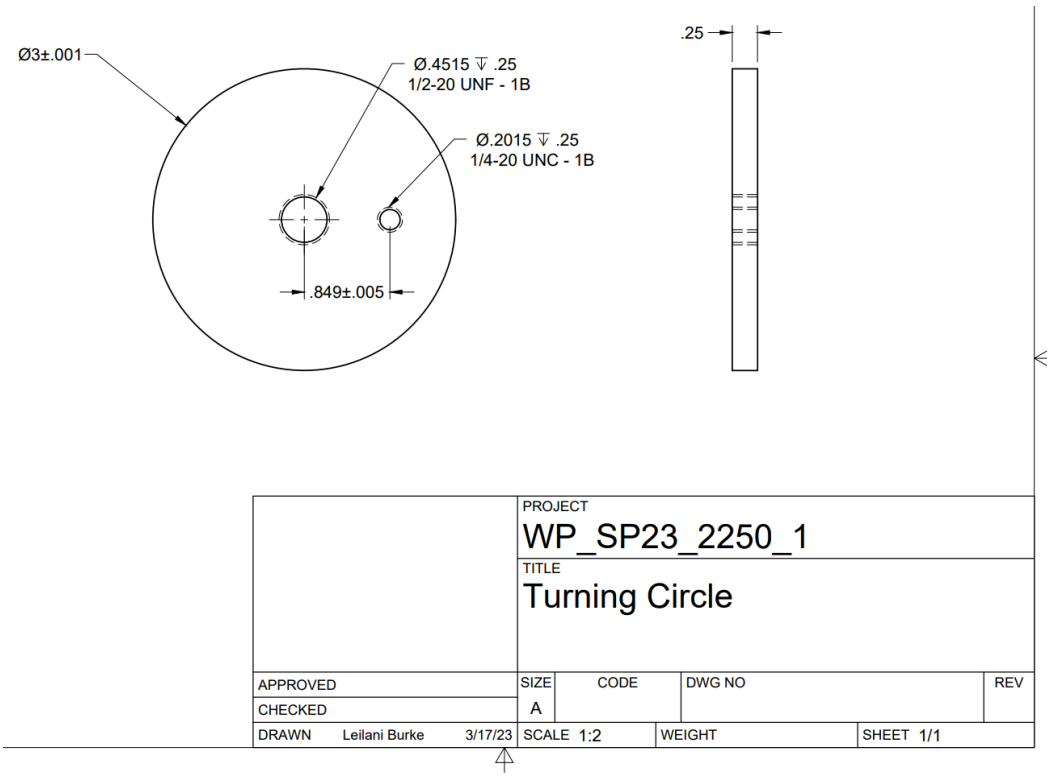


Figure A.8 Part drawing for the machined center turning circle

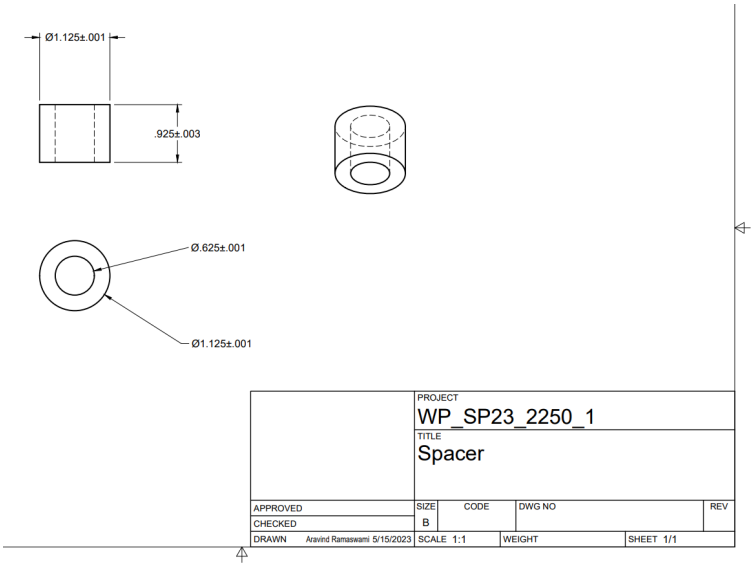


Figure A.9 Part drawing for the machined spacer

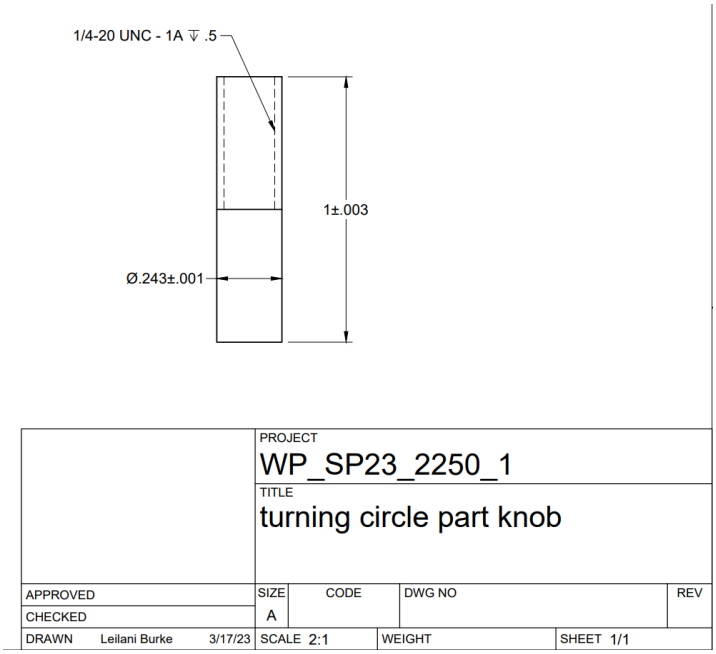


Figure A.10 Part drawing for the machined knob

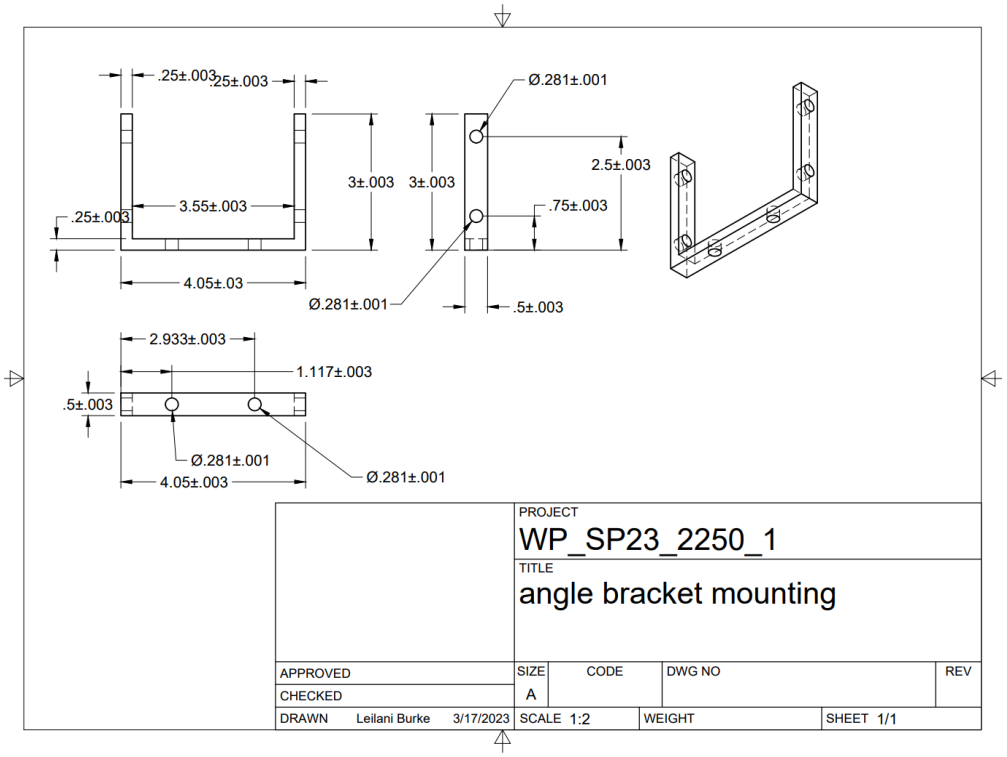


Figure A.11 Part drawing for the 3D-printed angle brackets

Appendix B: FEA Analysis

We ran FEA analysis on several key sections of the pump to determine if they would survive against the expected loads. A key point of the failure analysis was the pump heads and rods, as they were the most likely to fail.

B.1: Pump Heads

We ran a static stress analysis against the pump heads to see if they would deform or break. We supplied 75 psi to the pump heads to see if they would fail. From figures B.1, B.2, and B.3, we can see that the factor of safety is very high and the expected displacement is very low. This indicates that our design would survive the operational loads.

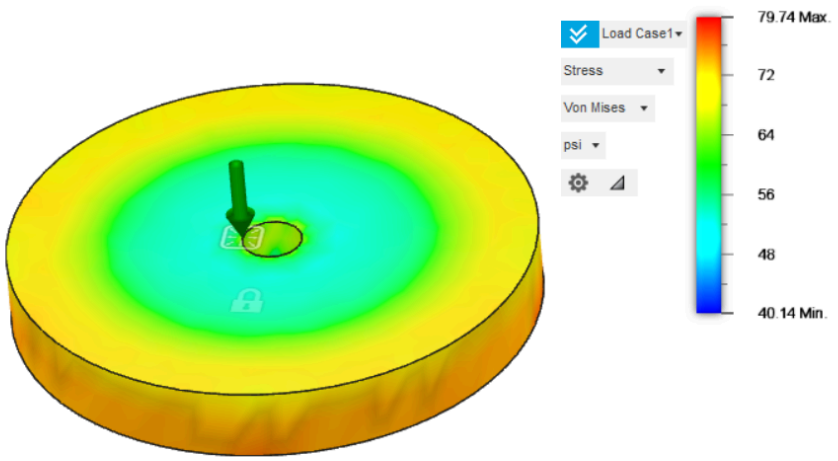


Figure B.1 Stress analysis for the pump head

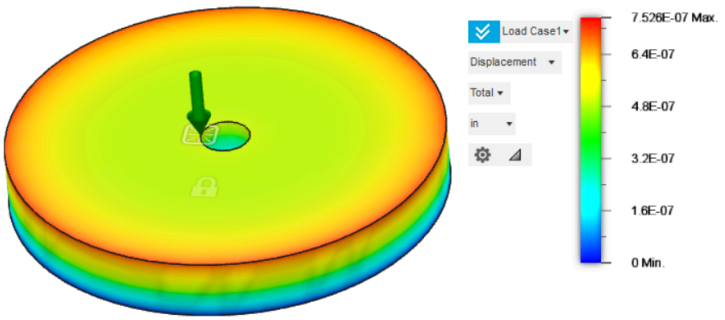


Figure B.2 Displacement analysis for the pump head

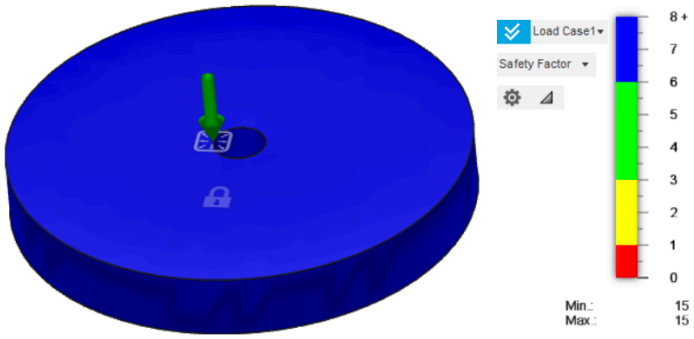


Figure B.3 FOS during the study

B.2: Rods

Similar to the pump heads, we ran a static displacement study to determine if the rods would break. We supplied 1000 lb to the rods. This is much greater than the actual operational load of the rods. This was done to ensure that the rods would not break. Figures B.4, B.5, and B.6 show the results. Although it is not as good as the previous part, 1000 lb is far greater than any load we would see during actual operation. Also, a safety factor of 1.7 still indicates that part would not yield under this load. Thus, this part would survive under normal loading conditions.

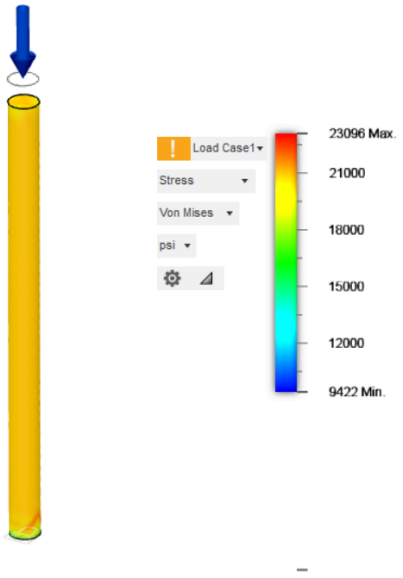


Figure B.4 Stress analysis for the rod

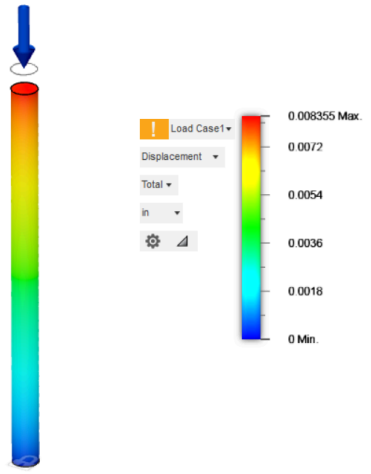


Figure B.5 Displacement analysis for the rod

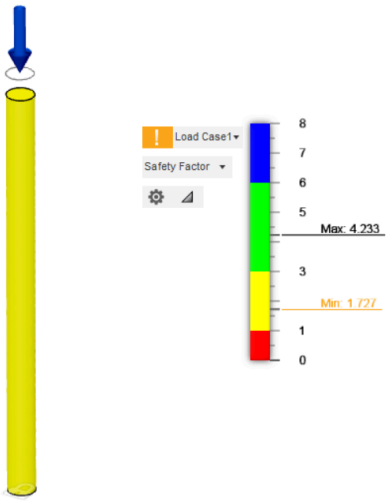


Figure B.6 FOS during the study

D. Sketches, Notebook drawings

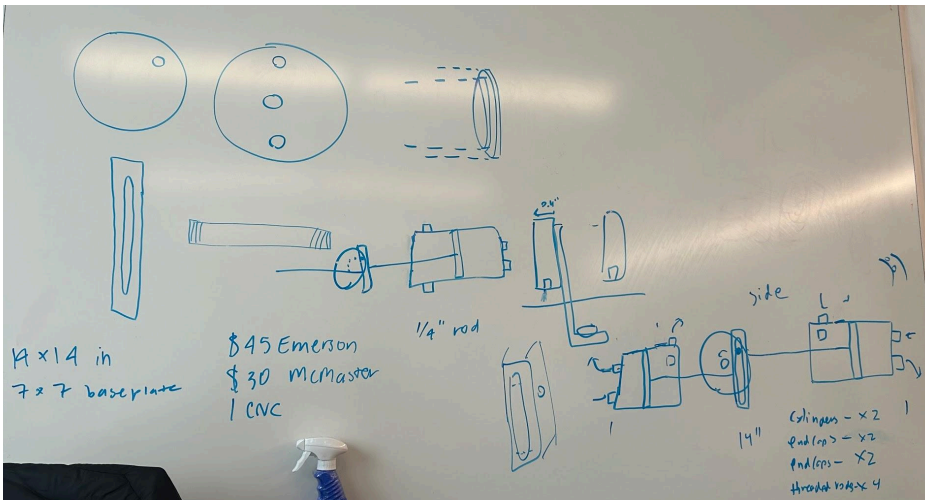


Figure 1.2 Initial white-board sketches during our first design meeting

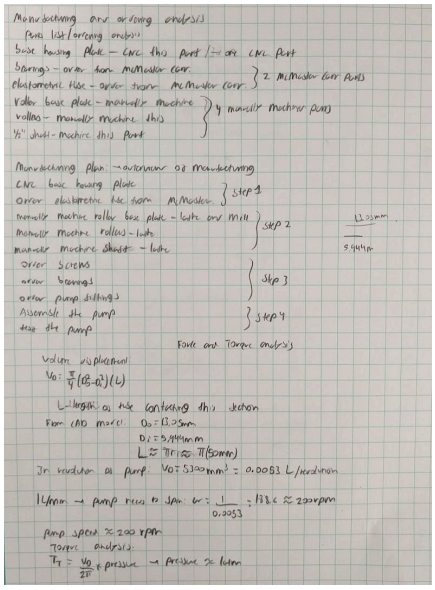


Figure 1.3 Notebook sketches

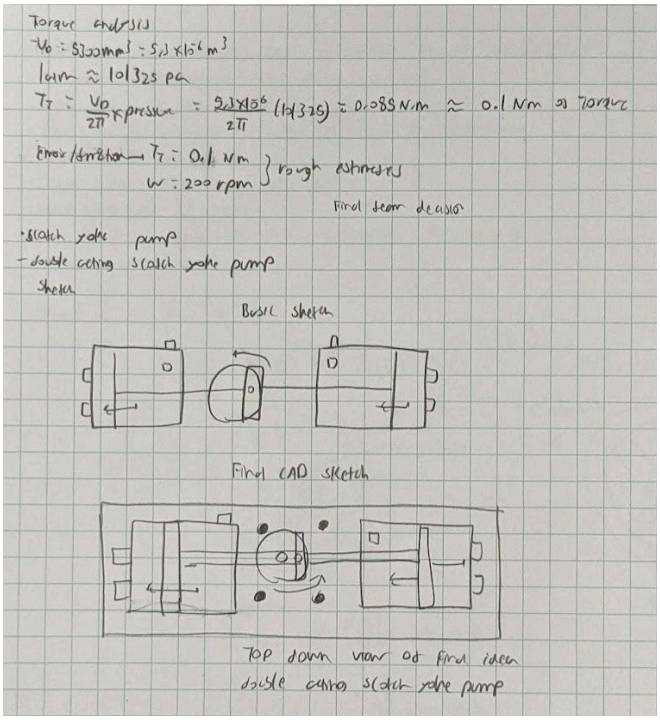


Figure 1.4 More notebook sketches



Figure 1.8 Cardboard prototype